Bholanath Bala

Date of Birth: 25/04/2002 | Gopalganj, Dhaka, Bangladesh | +8801934392790 |
ku.bhola16@gmail.com | linkedin.com/in/bhola-nath-bala | https://github.com/bhola16

RESEARCH INTERESTS

My research interests lie in **Artificial Intelligence, Machine Learning, Deep Learning, and Signal Processing**, particularly **transformer-based learning, time-series analysis, intelligent sensing, and data-driven intelligent systems**. I am interested in developing efficient and reliable AI methods for real-world sensing and information systems, with experience applying **Vision Transformers (ViT)** and **Time-Series Transformers (PatchTST)** to visual and vibration-signal data.

EDUCATION

Bachelor of Science (B.Sc.) in **Electronics and Communication Engineering (ECE)**.
Khulna University, Bangladesh
CGPA: 3.57/4.00 | **Graduated:** 2026

- **Undergraduate Thesis:** *A Comparative Study of Vision Transformers and Time-Series Transformers for Bearing Fault Diagnosis Using the PU Dataset.*
  - Developed transformer-based deep learning models for automated machinery fault diagnosis.
  - Implemented and evaluated **Vision Transformer (ViT)** on vibration-signal spectrograms and **PatchTST** on raw time-series signals.

RESEARCH PUBLICATIONS

- **Paul, A., Bala, B., Hasan, M. N., & Toma, R. N.,** "Vision Transformer-Based Classification of Bearing Faults from Vibration Signal Spectrograms," *IEEE-**ICCIT (2025).*** **DOI:** 10.1109/ICCIT68739.2025.11490241
- Maniruzzaman, M., Salman, R. I., Tawhid, K. G**., Bala, B.**, Hossain, M. M., & Ahsan, M. S., **"**A Novel Attention-Enhanced Hybrid Deep Learning Model for GI Disease Classification," *IEEE-PECCII (2026).* DOI: 10.1109/PECCII70991.2026.11661820

ACADEMIC PROJECTS

1. **Combating Unpaid Food Purchases Using OpenCV and Face Recognition**
   - Developed a computer vision system using **Python and OpenCV** to identify unpaid food purchases.
   - Applied **face recognition** techniques for automated user identification.
   - Designed the system with an emphasis on practical computer vision and intelligent automation in a canteen environment.
2. **Monitoring Environmental Conditions for Sundari Trees in the Sundarbans.**
   - Developed an IoT-based environmental monitoring system using sensors to monitor environmental conditions.
   - Designed web and mobile interfaces for real-time monitoring and automated alerts.
3. **University Certificate Automation System**
   - Developed an automated system for managing student certificate approval workflows with **real-time status tracking**.
   - Designed and implemented a **responsive, state-driven interface using React.js**.

TECHNICAL SKILLS

- **Programming:** Python, C, Java, JavaScript, TypeScript, MATLAB
- **DL & ML:** NumPy, Pandas, Matplotlib, PyTorch, Scikit-learn
- **Signal Processing & AI:** Time-Series Analysis, STFT, Representation Learning, Vision Transformers (ViT), Time-Series Transformers (PatchTST)
- **Computer Vision:** OpenCV, Face Recognition, Image Classification
- **Web Technologies:** HTML, CSS, Tailwind CSS, React.js, Next.js
- **Tools:** Git, GitHub, LaTeX

ACHIEVEMENTS

- **Champion**: **Research Category, Smart Bangladesh Seminar,** Khulna University, for the Sundarbans Environmental Monitoring Project.
- **2nd Runner-up: Machine Learning Poster Design Competition,** Khulna University.
- **Winner**: **Open Doors Russian Scholarship Project 2025 (Olympiad),** Applied Mathematics and Artificial Intelligence.

TRAINING & EXPERIENCE

- **Industrial Training in Data Science at Grameenphone, Bangladesh.**
  - Gained foundational knowledge in data analysis and machine learning workflows through industry-led training sessions

CERTIFICATIONS

- Programming for Everybody (Getting Started with Python), Coursera
- Intermediate Python, DataCamp
- Introduction to Machine Learning, Kaggle
- Introduction to SQL, DataCamp
- Introduction to Git, DataCamp

LANGUAGES

- **Bangla:** Native
- **English:** Intermediate

REFERENCES

1. **Dr. Rafia Nishat Toma**
   Associate Professor
   Electronics and Communication Engineering (ECE) Discipline
   Khulna University, Khulna-9208, Bangladesh
   Email: rafiatoma@ece.ku.ac.bd
   B.Sc. Thesis Supervisor
2. **Dr. Md. Maniruzzaman**
   Professor
   Electronics and Communication Engineering (ECE) Discipline
   Khulna University, Khulna-9208, Bangladesh
   Email: m_m_zaman@ece.ku.ac.bd